

T.POOJITHA (192412318)

ITA0609

Exp-14: Implement House price Prediction using appropriate machine learning algorithm

Import required libraries

```
from sklearn.datasets import fetch_california_housing
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.linear_model import LinearRegression
```

```
from sklearn.metrics import mean_squared_error, r2_score
```

Load the California Housing dataset

```
housing = fetch_california_housing()
```

```
X = housing.data
```

```
y = housing.target
```

Split the dataset into training and testing sets

```
X_train, X_test, y_train, y_test = train_test_split(
```

```
    X, y, test_size=0.2, random_state=42
```

```
)
```

Create the Linear Regression model

```
model = LinearRegression()
```

Train the model

```
model.fit(X_train, y_train)
```

Predict house prices

```
y_pred = model.predict(X_test)
```

Display results

```
print("Actual House Prices:")
```

```
print(y_test[:10])
```

```
print("\nPredicted House Prices:")
```

```
print(y_pred[:10])
```

Evaluate the model

```
print("\nMean Squared Error:", mean_squared_error(y_test, y_pred))
```

```
print("R2 Score:", r2_score(y_test, y_pred))
```

Output:

```
... Actual House Prices:
[0.477  0.458  5.00001 2.186  2.78  1.587  1.982  1.575  3.4
 4.466  ]

Predicted House Prices:
[0.71912284 1.76401657 2.70965883 2.83892593 2.60465725 2.01175367
 2.64550005 2.16875532 2.74074644 3.91561473]

Mean Squared Error: 0.5558915986952422
R2 Score: 0.5757877060324524
```